Selecting GAN-Synthetic Training Images by Committee Disagreement for Small-Data Satellite Road Segmentation

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Abstract

GAN-generated images are increasingly bundled with small annotated satellite datasets, but their value for road extraction is rarely measured cleanly. Auditing a public road-segmentation dataset (100 real images plus 1,003 Pix2Pix-generated pairs), we find only 106 distinct synthetic images and 75 distinct masks, each file repeated about twelve times. Under a protocol with a fixed real test set, five-fold real-only cross-validation, synthetic data confined to training, matched optimizer budgets and 15 paired replicates per condition, the synthetic images raise Attention U-Net test Dice from 0.806 to 0.839 once duplicates are removed; training on the 1,003 files as shipped is significantly worse than training on the 106 unique images, and validating on the pooled real–synthetic set inflates the apparent benefit about sevenfold. We then select synthetic images by the disagreement of a committee of real-trained models. On the duplicated set every score-based selection collapses to 26 unique images and loses to random sampling. After deduplication, the 50 most-disputed images match the full unique set, exceed a random 50 in 11 of 15 replicates, beat the 50 most-agreed images by 2.6 Dice points and give the lowest seed sensitivity of any condition. Deduplication, real-only evaluation and disagreement-based selection are recommended.

Keywords: road extraction; synthetic data; Pix2Pix; data duplication; evaluation leakage; query by committee; Attention U-Net

1 Introduction

Pixel-level road annotation is expensive, and conditional generative models offer a shortcut: Pix2Pix (Isola et al. 2017) translates a road mask into a plausible satellite image, so new mask layouts give new training pairs with perfect label alignment. Some public datasets, including the one studied here, bundle such GAN-generated training images with the real ones, and studies that use them, including the author's own earlier work, have reported large improvements. Practitioners nevertheless face three unanswered questions. How much of a synthetic set is genuinely new content? How much real-image benefit does it provide once the evaluation is clean? And can the useful synthetic images be selected in advance?

We answer all three for the Kaggle dataset *Satellite Images for Road Segmentation* (sanadalali n.d.), which ships 100 annotated satellite images and 1,003 Pix2Pix-generated pairs. The study is a corrected and extended version of a conference submission that reported a rise in Attention U-Net Dice from 0.81 to 0.96 when the synthetic images were added. Reviewers objected that the numbers came from a validation split drawn from the pooled real and synthetic images, and asked whether the synthetic set was "artificially repeated". Both objections turn out to be exactly right, and answering them properly produces findings that we believe are more useful than the original claim.

The contributions are as follows. (i) A file-level audit shows that the 1,003 synthetic pairs contain 106 distinct images and 75 distinct masks, each duplicated about twelve times. (ii) Under a leakage-free, compute-matched protocol with 15 paired replicates per condition, we measure the real benefit of the synthetic data on held-out real images (3.3 Dice points from the 106 unique images), show that training on the duplicated files as shipped is significantly worse than training on the unique images, and quantify the sevenfold inflation caused by validating on the pooled set. (iii) We show that query-by-committee selection of synthetic images fails on the duplicated set, because identical files receive identical scores and any ranking collapses to a few unique images, but succeeds after deduplication: the 50 most-disputed unique images match the full unique set and outperform random and agreement-based selections, with the lowest seed sensitivity of any condition. Code, the frozen split file, committee scores and all per-run results are released.[1]

2 Related work

U-Net (Ronneberger et al. 2015) and its variants dominate road extraction from overhead imagery (Zhang et al. 2018, Zhou et al. 2018, Batra et al. 2019). Attention U-Net (Oktay et al. 2018) adds additive attention gates to the skip connections, a modification of about 1% more parameters that suppresses background features; it is the architecture used throughout. GAN-based augmentation (Goodfellow et al. 2014, Isola et al. 2017) is widely used against annotation scarcity in remote sensing, but evaluation practice is inconsistent and gains are often measured on splits that include synthetic images. Duplicate and near-duplicate contamination has been documented in standard vision benchmarks, where copies shared between training and test sets inflate test accuracy (Barz and Denzler 2020); to our knowledge it has not been reported for a GAN-expanded remote-sensing training set, where its effect is different: it inflates the *apparent size* of the training set and defeats per-file selection rules. Query-by-committee (Seung et al. 1992, Settles 2009) selects unlabelled samples on which independently trained models disagree; its use for choosing among *synthetic* samples, whose labels are fixed by construction, has not been evaluated on a segmentation task with a real held-out test set.

3 Materials and methods

3.1 Data audit

The dataset provides 100 real RGB satellite images of 400×400 pixels with binary road masks (road fraction $20.3\% \pm 7.2\%$), 50 unlabelled test images that we do not use, and 1,003 416×416 image–mask pairs that the dataset page describes as generated with Pix2Pix. The provenance of the synthetic masks is not documented on the dataset page. Figure 1 shows real samples.

Hashing every synthetic file reveals that the set is heavily duplicated. There are **106 distinct synthetic images** and **75 distinct synthetic masks**: 45 masks were rendered once, 29 twice and one three times, and every resulting image was then copied, byte for byte, between 1 and 14 times (median 12). Every copy of an image carries the same mask. None of the 75 synthetic masks is pixel-identical to a real mask, or to a flipped or rotated real mask, after resizing the real masks to the synthetic resolution; we did not search for near-duplicates of real masks. The synthetic set therefore does not copy real test images

[1] [repository URL withheld for anonymous review; released on acceptance]

Table 1: Audit of the 1,003 synthetic image–mask pairs.

Synthetic files	1,003
Distinct images (byte-identical copies collapsed)	106
Distinct masks	75
Copies per distinct image: min / median / max	1 / 12 / 14
Distinct images per mask: one / two / three	45 / 29 / 1
Synthetic masks pixel-identical to a (flipped/rotated) real mask	0
Distinct committee-score pairs (fold 0)	93

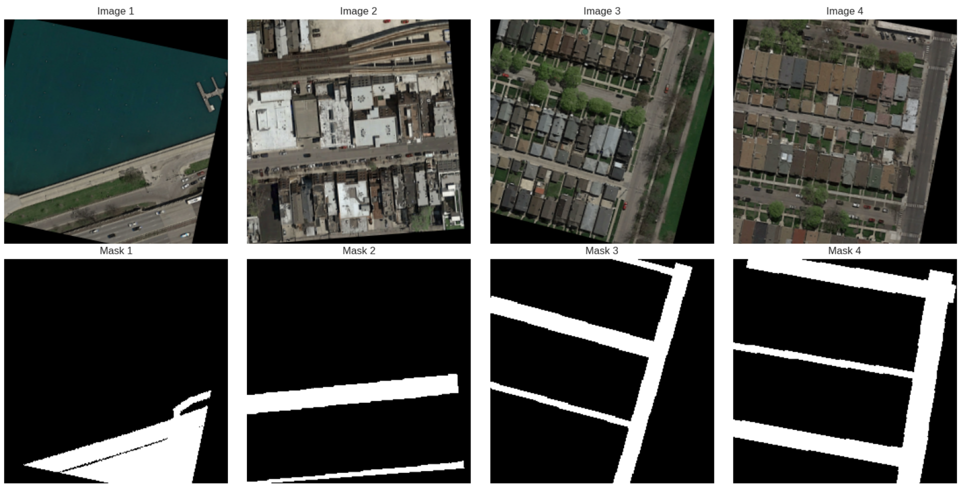

Figure 1: Real training samples spanning urban, suburban and rural scenes (top) and their binary road masks (bottom).

outright, but its effective size is about a tenth of its nominal size. Table 1 summarizes the audit. Two consequences follow for training: "1,003 synthetic images" is a twelve-fold oversampling of 106, and any per-file selection rule will select whole groups of identical copies.

3.2 Evaluation protocol

Real-only cross-validation. The 100 real images are sorted by road fraction and stratified into a fixed **test set of 20** and **five folds of 16**; the assignment is written to a split file shared by every experiment and released with the code. For fold f, the 16 images of fold f are the validation set, the other 64 are the real training set, and the 20 test images are untouched. Synthetic images are added to the training set only. Model selection uses the real validation fold; the test set is scored once per run with the selected weights.

Compute-matched training. Adding 1,003 files multiplies the updates per epoch by 17. To separate the effect of data from the effect of more optimization, every run trains for exactly 1,200 optimizer steps at batch size 8 (Adam, learning rate 10^{-4}, weight decay 10^{-5}, 100 warm-up steps then cosine decay to 10^{-6}) regardless of training-set size. Images are resized to 256×256 and normalized with ImageNet statistics; augmentation uses flips, 90° rotations, shift–scale–rotate, brightness/contrast, hue/saturation and Gaussian noise (Buslaev et al. 2020). The loss is binary cross-entropy plus Dice loss; mixed precision is used. The validation fold is evaluated every 100 steps and the best checkpoint by mean per-image Dice is kept. The Attention U-Net follows Oktay et al. (2018) with encoder

Table 2: Test results on the 20 held-out real images (mean ± SD over 15 fold×seed replicates). "Unique" is the mean number of distinct synthetic images present in the training set. Seed sensitivity is the SD across seeds averaged over folds. All runs use 64 real images and an identical 1,200-step budget.

Set	Condition	Files	Unique	Dice	IoU	Seed sens.
–	Real only	0	0	0.806 ± 0.015	0.698 ± 0.018	0.014
Duplicated	+ all	1,003	106	0.828 ± 0.010	0.726 ± 0.013	0.009
	+ random	250	81	0.832 ± 0.011	0.731 ± 0.016	0.012
	+ most-agreed	250	26	0.803 ± 0.007	0.702 ± 0.010	0.007
	+ most-disputed	250	26	0.798 ± 0.021	0.677 ± 0.027	0.017
Deduplicated	+ all-unique	106	106	$\mathbf{0.839 \pm 0.014}$	$\mathbf{0.741 \pm 0.018}$	0.013
	+ random	50	50	0.830 ± 0.013	0.733 ± 0.015	0.012
	+ most-agreed	50	50	0.812 ± 0.010	0.712 ± 0.011	0.009
	+ middle band	50	50	0.815 ± 0.013	0.716 ± 0.019	0.012
	+ most-disputed	50	50	$\mathbf{0.839 \pm 0.009}$	0.738 ± 0.012	**0.007**

widths 64–128–256–512, a 1,024-channel bottleneck, transposed-convolution upsampling and attention gates on all four skip connections (31.4 M parameters).

Replication and statistics. Every condition is trained for all five folds with three seeds (42, 123, 456), giving 15 replicates; seeds control initialization, data order and augmentation. We report mean per-image Dice and IoU on the 20 test images, the standard deviation across replicates, and the mean over folds of the standard deviation across seeds (seed sensitivity). Conditions are compared with the Wilcoxon signed-rank test and the paired t-test over the 15 matched (fold, seed) pairs.

3.3 Committee-based selection of synthetic images

For each fold, the three real-only models form a committee. Each synthetic image is passed through the committee, predictions are binarized at 0.5, and the *committee disagreement* is the mean pairwise $1-\text{Dice}$ between the three predictions. The committee is trained only on the real training folds, so the score never sees validation or test images. For diagnosis we also record each member's $1 - \text{Dice}$ against the synthetic mask (*label disagreement*) and the mask's road fraction. Identical files receive identical scores.

Two families of conditions are compared, all with the 64 real training images. On the **duplicated set** (files as shipped): *all* (1,003 files); *random* (250 files drawn uniformly, re-drawn per fold and seed); *most-agreed* (250 lowest-disagreement files, i.e. filtering suspected artefacts); and *most-disputed* (250 highest-disagreement files, i.e. uncertainty sampling). On the **deduplicated set** (one file per distinct image): *all-unique* (106); and *random*, *most-agreed*, *middle band* (the 50 images centred on the median score) and *most-disputed*, each with 50 unique images. Together with the real-only baseline this is 10 conditions ×15 replicates = 150 training runs plus five scoring passes, about 15 GPU-hours on a single T4.

4 Results

Synthetic data helps by about three Dice points, and the copies hurt. Table 2 and Figure 2 summarize the 150 runs. Training on the 106 unique synthetic images raises test Dice from 0.806 to 0.839 (+3.3 points, 13/15 wins, Table 3). Training on the 1,003 files

Table 3: Paired comparisons of test Dice over the 15 matched (fold, seed) replicates. Δ is the mean difference $a - b$ with 95% confidence interval; "wins" counts replicates in which a exceeds b. Numbers in parentheses are files (duplicated set) or unique images (deduplicated set, marked †).

a	b	Δ Dice	Wins	Wilcoxon p	Paired t p
Duplicated set					
random (250)	real only	$+0.026 \pm 0.010$	13/15	3.1×10^{-4}	8.4×10^{-5}
all (1,003)	real only	$+0.022 \pm 0.010$	14/15	1.2×10^{-3}	3.4×10^{-4}
all (1,003)	random (250)	-0.004 ± 0.005	3/15	0.064	0.14
most-agreed (250)	random (250)	-0.029 ± 0.006	0/15	6.1×10^{-5}	1.1×10^{-7}
most-disputed (250)	random (250)	-0.033 ± 0.011	1/15	1.2×10^{-4}	1.3×10^{-5}
Deduplicated set					
all-unique (106)†	all (1,003)	$+0.011 \pm 0.007$	13/15	6.7×10^{-3}	4.7×10^{-3}
all-unique (106)†	real only	$+0.033 \pm 0.014$	13/15	3.1×10^{-4}	1.3×10^{-4}
random (50)†	real only	$+0.024 \pm 0.011$	13/15	8.5×10^{-4}	2.8×10^{-4}
most-disputed (50)†	real only	$+0.033 \pm 0.007$	15/15	6.1×10^{-5}	7.1×10^{-8}
most-disputed (50)†	random (50)†	$+0.008 \pm 0.009$	11/15	0.048	0.070
most-disputed (50)†	all-unique (106)†	-0.001 ± 0.011	10/15	0.80	0.90
most-disputed (50)†	most-agreed (50)†	$+0.026 \pm 0.008$	14/15	1.2×10^{-4}	8.9×10^{-6}
most-agreed (50)†	random (50)†	-0.018 ± 0.008	2/15	6.1×10^{-4}	3.8×10^{-4}
middle band (50)†	random (50)†	-0.015 ± 0.011	4/15	0.018	0.010

as shipped reaches only 0.828: the same content with twelve-fold repetition is significantly worse than the unique images under an identical budget ($\Delta = -0.011$, 13/15, $p < 0.01$), and no better than a random 250-file subset that happens to contain 81 unique images. Synthetic data also reduces seed sensitivity, from 0.014 to 0.009–0.013.

Evaluation on the pooled set inflates the benefit sevenfold. Under the earlier pooled-validation protocol, the same architecture and data showed a gain of 15.1 Dice points from synthetic expansion (mean validation Dice 0.810 to 0.961 over three seeds). Under the real-only protocol the gain is 2.2 points for the same 1,003 files. Two mechanisms combine: synthetic images are easier than real ones, having perfect label alignment and no annotation noise, and because each synthetic image appears twelve times, a validation split drawn from the pool almost always contains copies of images that are also in the training split, so the pooled protocol partly measures memorization.

On the duplicated set, committee selection collapses diversity and loses to random. Because identical files receive identical scores, ranking by committee disagreement selects whole groups of copies: the 250 most-agreed files contain 26 unique images (range 24–27) and so do the 250 most-disputed, against 81 for a random draw. Both selections lose to random in nearly every replicate (mean paired difference $\Delta = -0.029$ and -0.033 Dice, 0/15 and 1/15 wins, $p < 10^{-3}$) and neither differs significantly from using no synthetic data at all (Figure 3, left). This is the realistic setting for a practitioner who downloads a GAN-expanded dataset and applies a per-image score, and the lesson is that any such score must be applied after deduplication.

After deduplication, disagreement selection works. With one file per unique image, the ordering reverses (Figure 2, bottom; Figure 3, right). The 50 most-disputed unique images reach Dice 0.839, indistinguishable from the full set of 106 ($\Delta = -0.001$, $p = 0.8$) while using less than half of it, above a random 50 in 11 of 15 replicates ($\Delta = +0.008$;

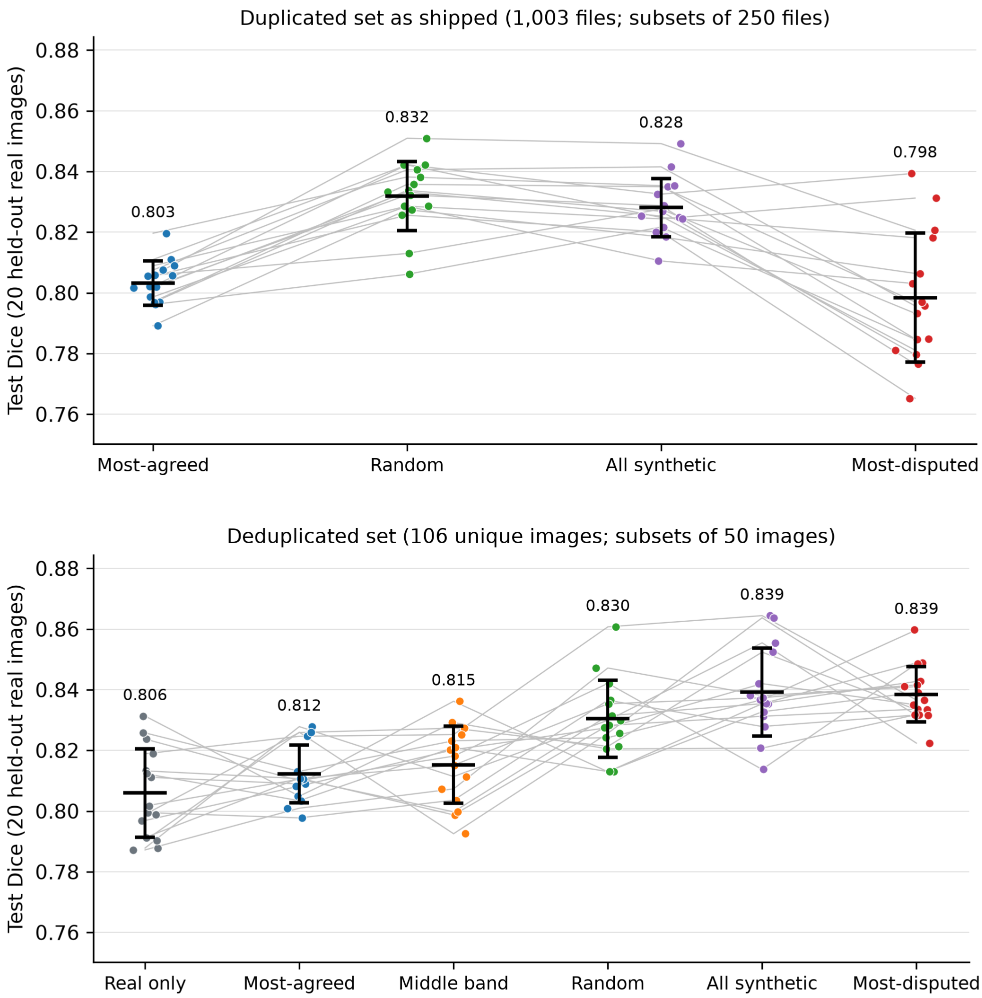

Figure 2: Test Dice of every replicate per condition on the duplicated set as shipped (top) and after deduplication (bottom). Grey lines join the same (fold, seed) across conditions; black bars show mean ± SD. Both panels share the vertical scale.

Wilcoxon $p = 0.048$, paired t $p = 0.07$), above the 50 most-agreed images by 2.6 points (14/15, $p < 10^{-3}$), and above real-only data in every replicate. They also give the lowest seed sensitivity of any condition (0.007, half that of real-only training). The 50 most-agreed images and the middle band are both significantly worse than random and add little over real data alone. This is the classical query-by-committee ordering (Seung et al. 1992): the images on which independently trained models disagree are the informative ones.

What committee disagreement measures. Figure 4 and the per-fold diagnostics show that committee disagreement is strongly correlated with label disagreement (Spearman rank correlation coefficient $\rho = 0.86$–0.89 across folds): the images the committee disputes are those whose masks contain road structure that real-trained models do not yet predict. Given the deduplicated results, this is a signal of informativeness rather than of label noise. Disagreement is also negatively correlated with road density ($\rho = -0.25$ to -0.47), so the most-disputed subset under-represents dense-road scenes (mean road

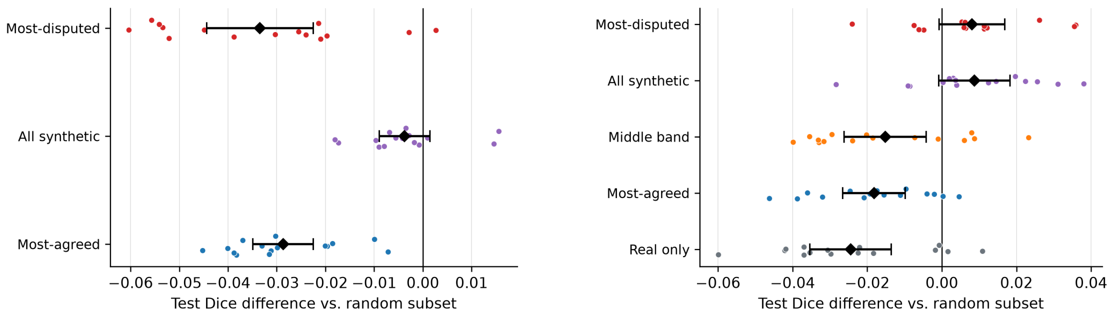

Figure 3: Paired differences in test Dice relative to the random subset of the same family: 250 files on the duplicated set (left) and 50 unique images on the deduplicated set (right). Diamonds and bars show the mean and 95% confidence interval over 15 replicates.

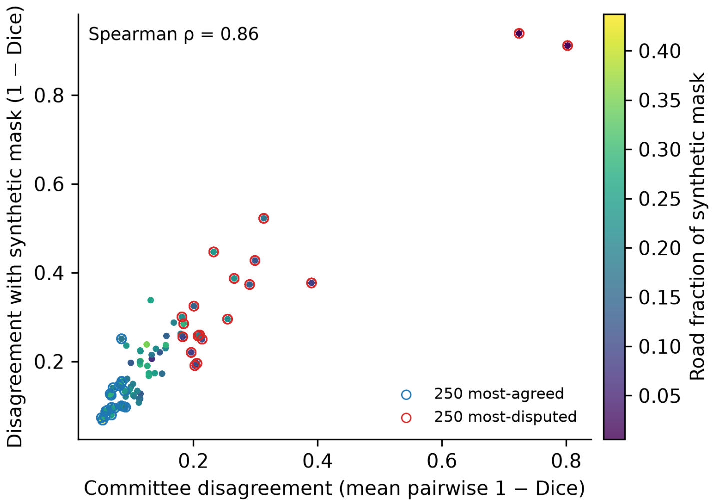

Figure 4: Committee disagreement versus disagreement with the synthetic mask for the synthetic set (fold-0 committee), coloured by road fraction. Identical files overplot, so the 1,003 files appear as 93 points. Circles mark the 250 most-agreed (blue) and most-disputed (red) files.

fraction 0.14–0.17 versus 0.20 for real data) and the most-agreed subset over-represents them (0.22–0.25); the most-disputed images win despite this shift. The ranking is a stable property of the images: on average 67% of the 250 most-agreed files coincide across committees trained on different real folds. Why, then, did the most-disputed selection fail on the duplicated set? There it consisted of 26 unique images repeated about ten times each, and the deduplicated experiment shows separately that repetition of a fixed set is harmful under a matched budget (all-unique versus all). Narrow content plus heavy repetition, not the disagreement criterion, explains the earlier loss.

5 Discussion

Three practical rules follow. *Audit before use*: a GAN-expanded dataset should be hashed for exact and near duplicates, because nominal size can overstate content by an order of magnitude, repetition itself costs accuracy under a fixed budget, and per-file selection or weighting rules behave pathologically on duplicated sets. *Evaluate on real images only*: hold out real images before anything synthetic is added, cross-validate on real images, and match the optimization budget across conditions; otherwise gains are inflated many-fold, as the author's own earlier result shows. *Select by disagreement, after deduplication*: a committee of models trained on the real data, which costs nothing beyond the seeds one should be running anyway, identifies a subset of synthetic images that delivers the full benefit of the set with less than half the images and the most stable training.

The modest size of the real benefit, about three Dice points from 106 unique synthetic images derived from 75 masks, bounds what Pix2Pix expansion offers here. How the 75 synthetic masks were produced is not documented; if, as their layouts suggest, they were derived from the 100 real masks, the generator's effective diversity is limited by construction. Whether a diffusion model or a larger pool of independent masks would help more is an open question that our protocol makes cheap to answer, and disagreement-based selection is most useful precisely when the candidate pool is large.

Limitations. The study uses one benchmark of 100 real images, one generator whose synthetic masks were produced by a third party, one architecture and two subset sizes. Absolute Dice differs by fold (Figure 2), reflecting the small test set; our conclusions rest on paired within-fold comparisons, which are robust to this. The advantage of disagreement selection over random selection at equal size is modest (+0.8 Dice points) and significant by the Wilcoxon test but only marginally by the paired t-test; its advantages over agreement-based selection, over real-only training, and in seed sensitivity are large and unambiguous. Baseline U-Net was not re-evaluated under the new protocol. Duplicates were detected at the byte level; near-duplicate renderings of the same mask (29 masks have two) were kept as distinct images.

6 Conclusion

A public GAN-expanded road dataset advertised as 1,003 synthetic images contains 106, each repeated twelve times. Under a leakage-free, compute-matched, 15-replicate protocol those images improve Attention U-Net road segmentation on real held-out imagery by 3.3 Dice points, the repeated copies reduce that benefit, and validating on the pooled set inflates it sevenfold. Committee-disagreement selection of synthetic images fails on the duplicated set, where it collapses to a handful of unique images, but after deduplication the 50 most-disputed images match the full unique set, beat random and agreement-based selection, and halve seed sensitivity. Deduplicating synthetic sets, evaluating on real images only, and selecting synthetic images by committee disagreement should be default practice for GAN-expanded remote-sensing datasets.

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Disclosure statement

The author reports there are no competing interests to declare.

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Declaration of generative AI use

Generative AI (Claude, Anthropic) was used to assist in writing the experimental pipeline code and in language editing of this manuscript. It was not used to generate data, results or references. The study design, execution, analysis and conclusions are the author's own, and all reported numbers were produced by the released code.

Data availability statement

The dataset is publicly available on Kaggle (sanadalali n.d.). Code, the frozen split file, committee scores and all per-run result files that support the findings are openly available at [repository URL withheld for anonymous review; released on acceptance].

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